

CHATEAURROUX AP, France

WMO: 073540

Lat: 46.870N

Lon: 1.741E

Elev: 157

StdP: 99.45

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 34048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-3.5	-8.4	1.9	-3.9	-6.2	2.3	-1.9	14.0	10.8	12.4	9.7	3.4	80	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.7	20.3	30.3	19.7	28.2	18.9	21.5	29.6	20.6	28.3	19.8	26.5	3.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.0	18.2	13.3	23.0	17.4	12.7	22.2	63.1	29.7	60.0	28.3	57.2	26.7	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.4	8.3	-7.8	36.1	2.9	2.2	-9.9	37.7	-11.6	39.0	-13.2	40.2	-15.3	41.8		
			DB													
			WB	-8.1	23.3	2.9	0.8	-10.2	23.9	-11.9	24.4	-13.5	24.9	-15.6	25.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	4.5	5.2	8.1	10.7	14.3	18.1	20.2	20.0	16.4	12.9	7.8	5.0
	DBStd	6.65	4.10	4.01	3.42	3.25	3.26	3.49	3.17	3.30	3.02	3.63	3.64	4.14
	HDD10.0	680	173	138	77	29	4	0	0	0	1	16	83	159
	HDD18.3	2523	428	368	316	230	130	46	16	18	73	171	315	413
	CDD10.0	1404	3	3	19	50	138	244	316	309	193	105	18	5
	CDD18.3	204	0	0	0	0	6	39	74	69	15	2	0	0
	CDH23.3	2269	0	0	0	15	86	445	795	731	178	21	0	0
	CDH26.7	811	0	0	0	1	11	151	300	302	45	2	0	0
Wind	WSAvg	3.8	4.4	4.4	4.3	3.9	3.5	3.4	3.6	3.4	3.4	3.5	3.9	4.3
Precipitation	PrecAvg	781	66	56	50	70	74	57	60	61	62	77	75	72
	PrecMax	994	148	100	137	194	173	108	135	153	170	126	146	162
	PrecMin	558	12	7	6	12	23	13	6	10	19	37	27	16
	PrecStd	103	32	28	34	42	34	27	30	37	41	28	31	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.9	21.5	25.3	28.3	34.2	35.0	37.0	30.6	25.9	18.8	15.2
		MCWB	11.1	10.6	13.2	16.6	19.0	21.2	20.6	20.8	18.6	17.5	14.2	11.7
	2%	DB	12.7	14.2	18.8	22.5	25.9	30.5	32.2	33.0	27.7	23.0	16.2	13.3
		MCWB	10.7	9.9	11.8	14.7	17.6	20.1	20.5	20.0	17.9	16.9	13.5	11.3
	5%	DB	11.4	12.2	16.3	19.9	23.8	28.0	30.0	30.1	25.2	20.6	14.6	12.1
		MCWB	9.8	9.5	11.0	13.5	16.8	19.3	19.6	19.6	17.3	15.9	12.5	10.6
	10%	DB	10.2	10.8	14.2	17.4	21.5	25.7	27.9	27.4	23.0	18.5	13.2	10.9
		MCWB	8.8	8.8	10.2	12.1	15.7	18.3	18.8	18.6	16.3	15.1	11.5	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.0	14.1	17.5	20.7	23.0	22.4	22.4	20.7	18.6	15.5	12.8
		MCDB	13.2	15.7	19.8	23.9	26.1	30.6	31.0	30.9	26.3	23.2	17.4	14.0
	2%	WB	11.1	10.7	12.7	15.4	18.8	21.1	21.4	21.2	19.2	17.5	14.0	11.7
		MCDB	12.2	12.7	16.7	21.1	24.2	28.4	29.7	30.4	25.0	21.2	15.9	13.0
	5%	WB	10.0	9.9	11.7	13.9	17.3	20.1	20.5	20.3	18.1	16.5	12.9	10.7
		MCDB	11.2	11.8	15.2	18.8	22.4	26.5	28.1	28.2	23.4	19.7	14.4	11.9
	10%	WB	8.9	9.0	10.7	12.6	16.1	19.0	19.6	19.3	17.1	15.5	11.6	9.7
		MCDB	10.0	10.7	13.7	16.6	20.6	24.3	26.4	25.8	21.7	18.2	13.1	10.9
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.5	9.5	10.7	10.6	11.2	11.7	11.6	11.1	9.1	6.8	6.0
		MCDBR	6.8	10.2	13.8	15.4	14.6	15.5	16.2	16.7	15.2	12.3	9.1	7.2
		MCWBR	5.1	6.6	8.0	8.5	7.6	6.8	6.2	6.3	7.0	6.8	6.3	5.6
	5% WB	MCDBR	6.0	8.5	11.3	13.8	13.1	14.0	14.1	13.9	12.1	10.6	8.2	6.8
		MCWBR	5.0	6.0	7.0	8.0	7.4	6.9	5.8	5.9	6.1	6.2	6.2	5.5
Clear-Sky Solar Irradiance	taub	0.316	0.336	0.364	0.395	0.397	0.410	0.402	0.394	0.362	0.358	0.330	0.313	
	taud	2.456	2.399	2.325	2.238	2.281	2.283	2.301	2.340	2.408	2.429	2.456	2.448	
	Ebn at Noon	774	829	855	858	868	856	858	851	849	792	748	731	
	Edh at Noon	68	87	108	129	128	129	125	116	99	84	67	61	
All-Sky Solar Radiation	RadAvg	1.09	1.93	3.15	4.44	5.20	5.93	5.77	5.08	3.96	2.40	1.32	0.94	
	RadStd	0.12	0.27	0.38	0.55	0.47	0.57	0.39	0.45	0.27	0.19	0.14	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page